

# Team Performance Of NFL Offenses From 2019-2021 NFL Seasons

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# Introduction

The purpose of our project was to analyze NFL Offenses using R and create visualizations that would help us determine important factors in winning at the top level of football. The dataset that we utilized was the Advanced Sports Analytics' NFL pass, rush, and receive data for the last three NFL seasons (2019 - 2021). This data we chose has 19,973 player performances with 69 variables over 1640 NFL games, which includes stats on every player in each game, the outcomes of each of those games, and sports betting values for games.

Football is an ingrained part of American culture and we found interest in learning about some of the deeper elements of the sport, especially what interesting findings we could discover looking through the myriad of statistics that can be gathered. We found it intriguing to look into NFL statistics as well for the fantasy and sports betting aspects of the game as well which has grown greatly over the last few years. We also decided to hone our focus on the offensive game, because while the defense does have interesting information to investigate the offense incorporates more diverse positions and a larger breadth of statistics to analyze.

For individual player stats we will look at their positions and see how the positions compare in terms of their fantasy points. We will also investigate if there is a drop in performance through fantasy points for players as the season progresses with an interest in player endurance throughout the season. For game results we will look into how heavy losses from the previous week will affect a team's performance, in addition to that we will explore the home field advantage and if it has an effect on the underdog team. For sports betting we also looked at which teams had the most consistent performances that would warrant betting on them more than the average team, using the over/under (A betting statistic that comprises the sum of both teams scores at the end of the game) and the spread (how much the team favored to win is expected to win by).

## Data Cleaning

Given the vast amount of variables in our database, we subsetting the data to about 30 variables. We also wanted to analyze NFL games in the given timeframe, so we created a separate database that combined rows into their respective game data for each team in the game. The diversity of our questions also meant the data was further subsetting into different databases respective to the questions. We will provide the raw data as well as the subsets used, which include the following:

team\_game\_performance.csv: The database containing all the team performances.

change\_from\_blowout.csv: The database that shows the change in variables from a blowout game to a game after.

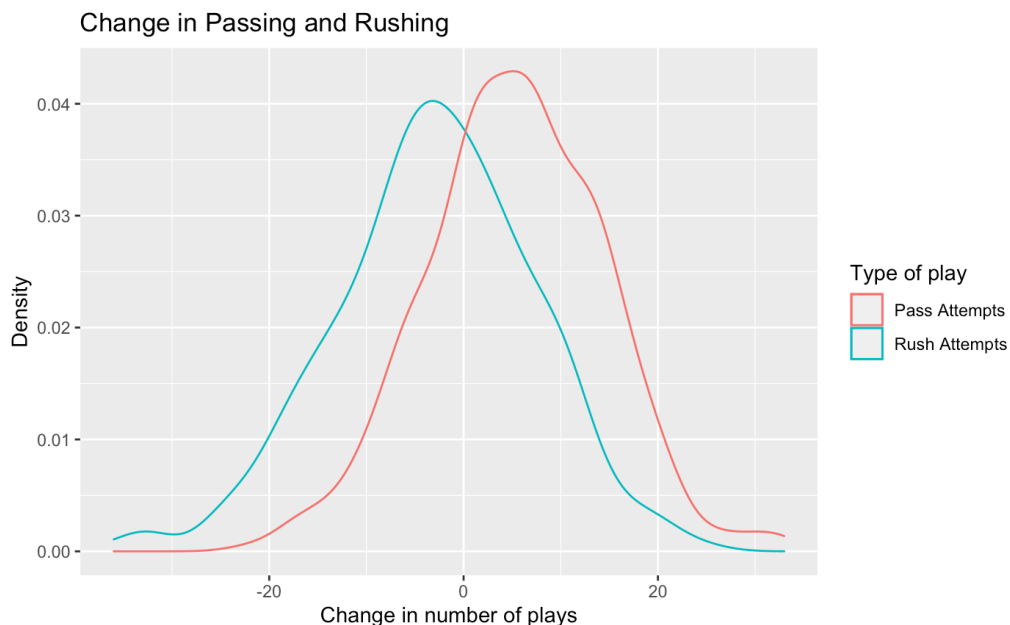
We also created several variables, including: if a team covered the spread, if a game covered the over/under, how many points a team won or lost by, if a team got blown out, if a team was an underdog, and how many scores a team won or lost by.

# Questions

## 1. How much does a loss by 10 or more points impact a team's performance for the next game?

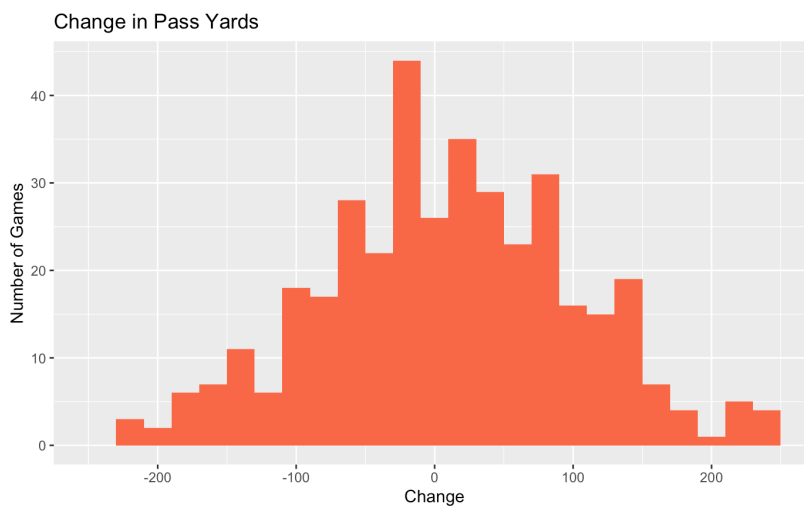
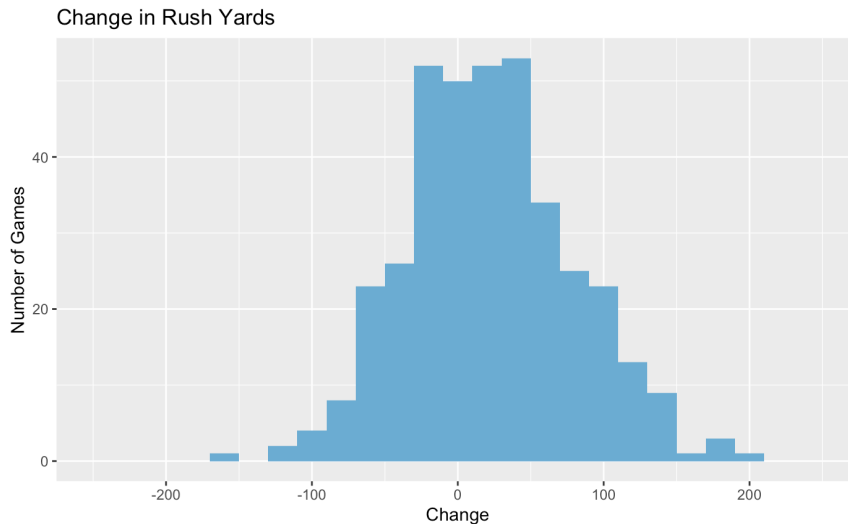
A game is considered a blowout if a team loses by 10 or more points. We wanted to look at how a team will be affected after losing in a blowout. There are many variables that could be in play, such as morale, game adjustments, and motivation. We focused on the more quantifiable variables, such as the change in amount of yards gained through different plays, the change in the amount of plays played, and the change in scoring plays or turnovers. By looking at these changes, we can uncover what teams at the highest level of football adjust after a huge loss.

The first variable we looked at was the change in types of plays. After every game, a team usually makes adjustments on how many of each type of play they run. This could be because of poor performance or simply because of a different opponent's defense. There are two types of plays: rushing, where the player with the ball runs, and passing, where the player with the ball throws the ball forward. We looked at the change from the blowout and the game after.



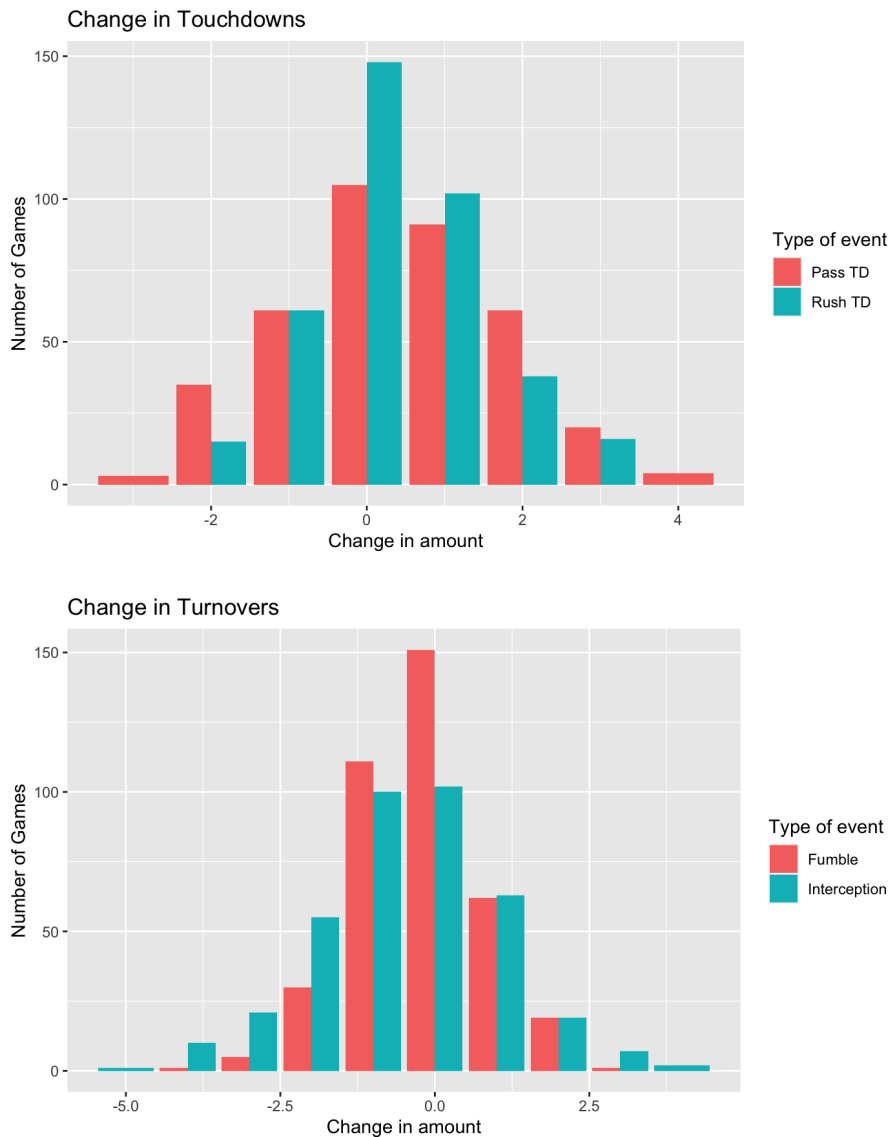
We found that on average, teams after losing a blowout will run more pass plays and rush plays than before. This may be due to the fact that a pass play's potential is higher as players without the ball aren't as targeted. We also found that the spread was pretty high, so there may be some variation of this variable.

To confirm this, we then looked at the change in pass yards and rush yards.



We found that the change in pass yards had a much higher spread than the change in rush yards. This may be due to the fact that more passing plays were attempted with a variation in how successful they were. Both the change in passing yards and the change in rushing yards skewed slightly positive, which indicates that any adjustments may have been successful or the team trained relatively harder.

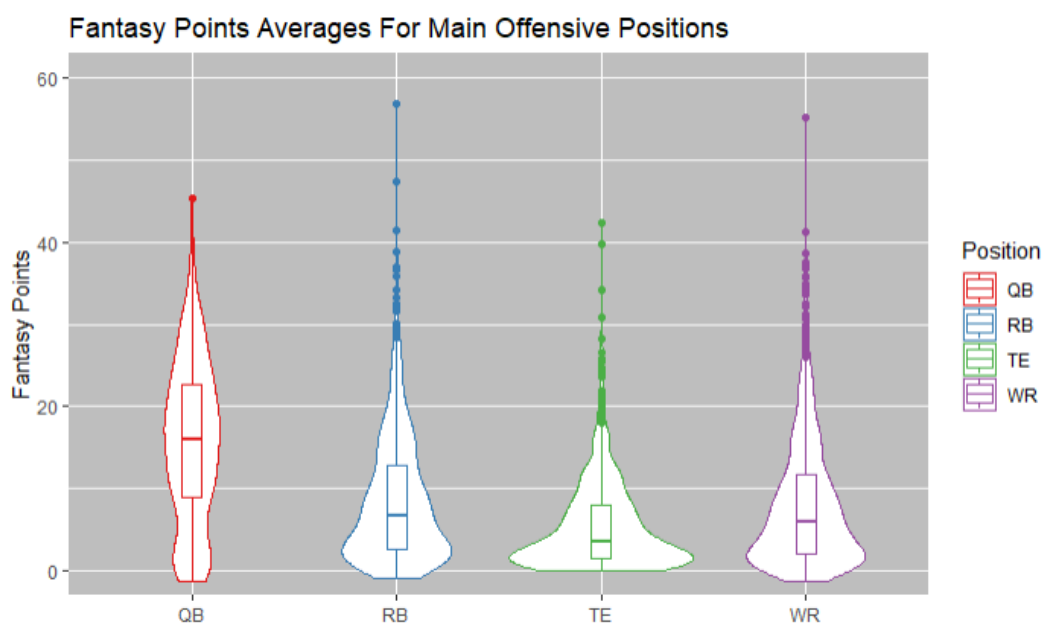
The last metric we looked at was the change in significant events. There are two types of significant events: touchdowns, which is when the offense scores in the end zone, and turnovers, when the other team's defense makes the offense lose possession of the ball. Either event can be the result of a pass or rush play.



Regarding touchdowns, we found that the change in both passing and rushing touchdowns both skewed positively. This means that any factors that made the team perform better, such as more motivation or easier opponents, were more likely. Similarly, we found that teams turned the ball over less, as indicated with the change in both fumbles (rushing turnovers) and passing turnovers (interceptions) skewed negatively. However, the change in interceptions tended to skew more than the change in fumbles.

Alongside the increase in passing plays and yards, this may mean that teams after a blowout went for safer, short throws.

## 2. Are QBs, RBs, TEs, or WRs more valuable in fantasy football?



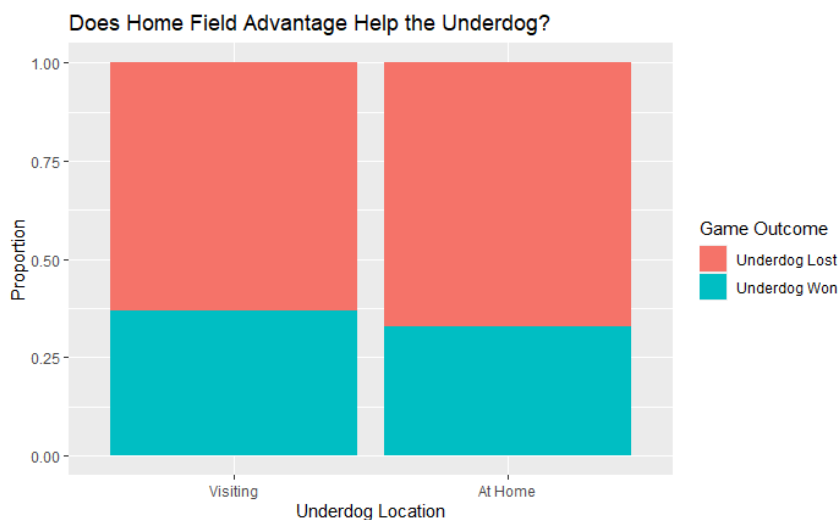
Fantasy football is a game in which players can build rosters and earn points from those teams. Points are scored through the standard method through points per reception (PPR), which means players will score a point for every catch they make. Rosters typically include 1 quarterback (QB), 2 running backs (RB), 2 wide receivers (WR), 1 tight end (TE), 1 flex (RB/TE/WR), 1 defense/special teams (D/ST), 1 kicker (K) and 7 bench spots.

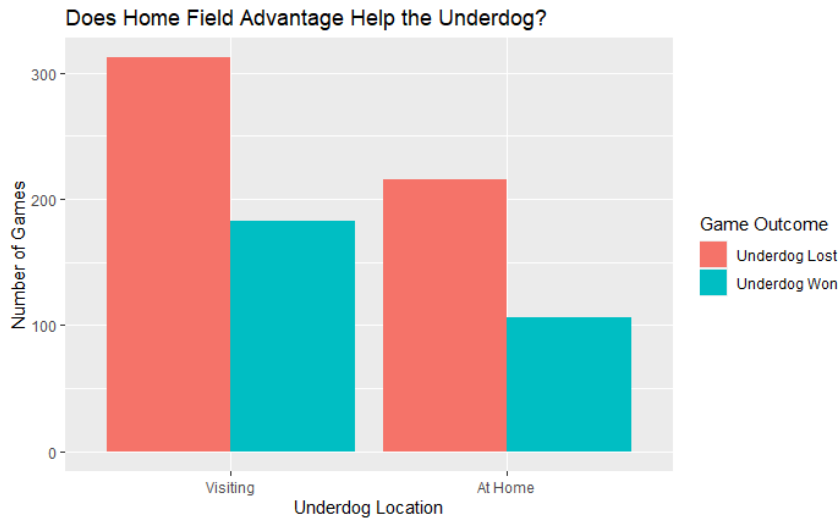
This question seems straightforward, but there is statistical information we make out of this. Before we can make the graph, we took the averages of all of the fantasy points of each player since we felt that the points earned within each fantasy football brand that was in the dataset would be too redundant. We initially made a box plot to compare the averages among the main offensive positions. But after looking at the plot, we felt that it did not tell how many players got a specific amount of points. Therefore, we fitted a box

plot within a violin plot for each position to better illustrate the spread of the averages amongst the players.

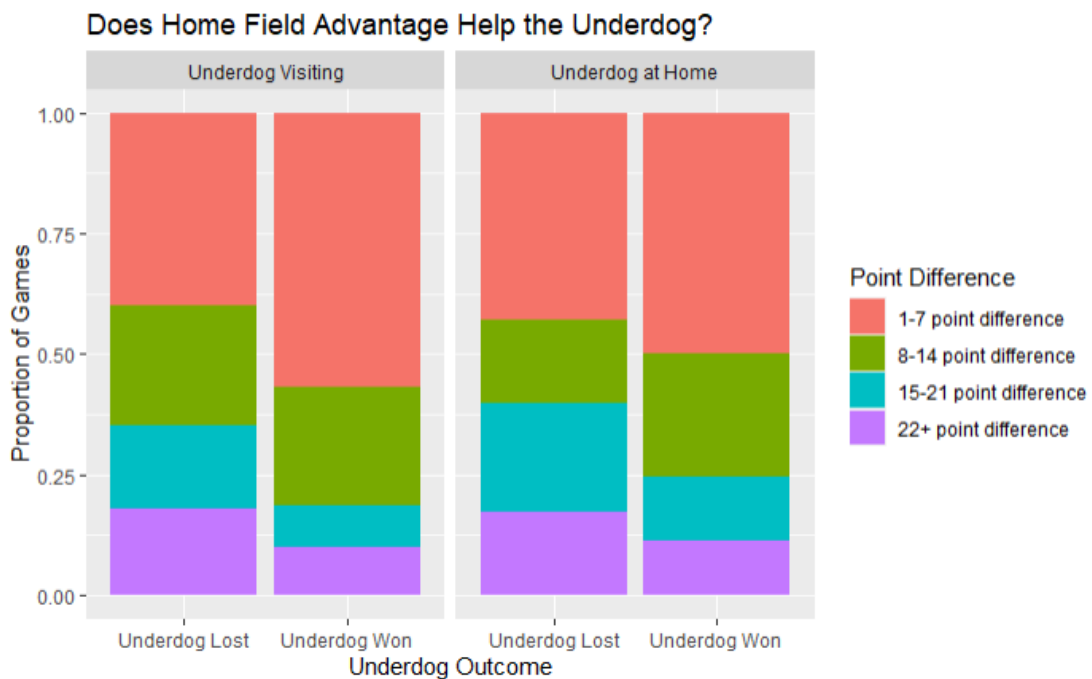
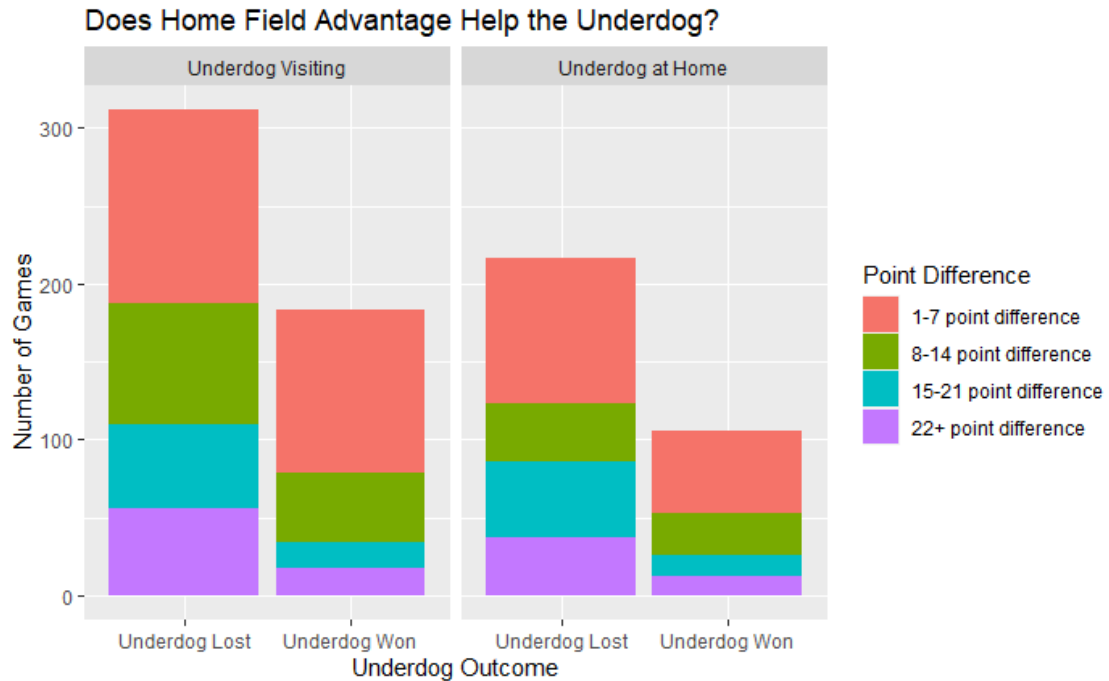
There are some interesting results coming from this graph. First of all, all of the plots seem to be skewed towards the bottom. This is because we only considered players who contributed a part in the game, whether it was catching the ball or making a run. If we had been more strict on our considerations, we could leave out some key data points. In addition, the quarterbacks usually had the most fantasy points on average compared to the running backs, tight ends, and wide receivers. The reason for this is that the quarterback is usually the leader of the offense as he is responsible for throwing the ball or handing it off for a rush attempt. On the other hand, the tight ends had the least fantasy points on average as they are mostly used for blocking. What is most interesting about this graph is the number of outliers for all of the positions. As most of the players we considered had low fantasy points, some had points on the other end of the spectrum as they put forth impressive performances.

### 3. How helpful is the home field advantage for underdogs?





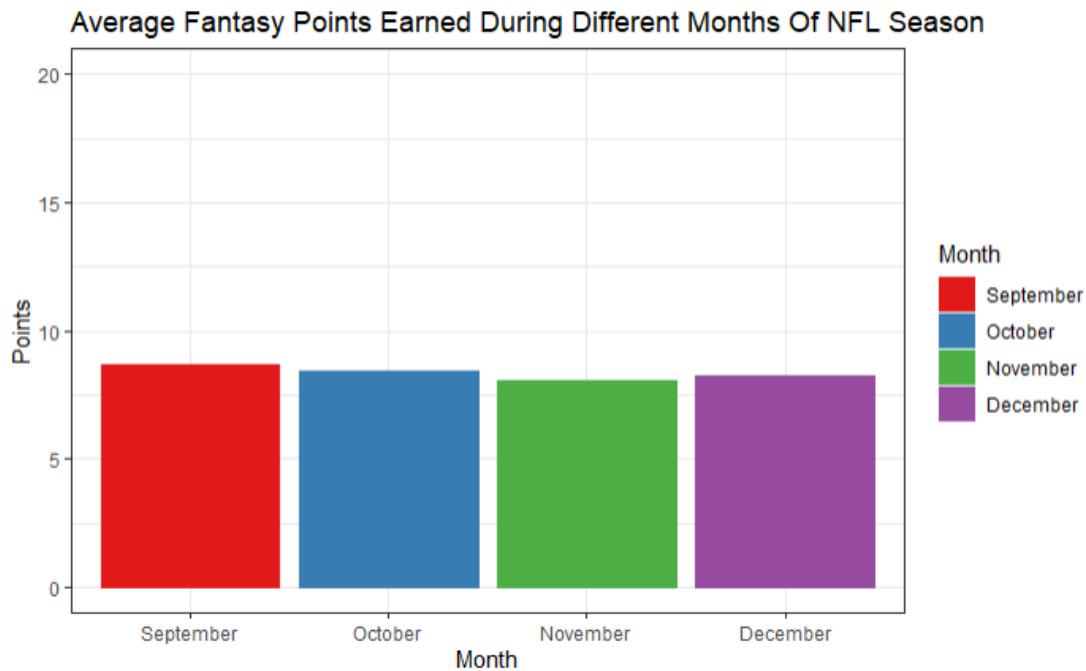
We initially created a bar plot with the proportion of games that underdogs would win or lose depending on if they were at home. This plot revealed the unexpected result that underdogs win less than 40% of their home games, which is odd because the home field advantage is understood as a benefit. This clearly was an odd result so we decided to look at the plot again, but instead of the proportion of games we looked at the number of games played. This showed another interesting result, the number of games that underdogs play at home is significantly less than the ones they play while visiting. This makes sense when factoring in the home field advantage, if there is a significant benefit when playing at home then it would follow that the home team would be the favorite more often than not. Which means that in games with evenly matched teams, the visiting team is more likely to be the underdog and the inverse of this would be true as well. If a game were to be wildly unfavorable for a given team, then the home field advantage wouldn't give them enough of an edge to be the favorite even if they were playing at home.



While this was an interesting result already, we didn't believe it was sufficient and decided to look into the point differences between the teams in these games. These graphs showed some information that was a little more expected like when underdogs win games they win by smaller margins regardless of if they are at home or visiting. Some interesting results did arise though, while visiting underdogs play proportionately more

close games (within 14 points) than they do at home, and the inverse is true as well while at home underdogs play proportionately more blowouts than they do when visiting. This backs up what was previously stated from the past graphs, and adds to the understanding of the home field advantage and its significance in NFL games. From these graphs we can reasonably conclude that the home field advantage does have a slight sway in the deciding of who is the favorite/underdog for any given matchup and is likely to add a significant difference to the score in these games.

#### 4. Do players experience burnout and decrease in performance over the course of the season?



For burnout and decrease in performance, we initially looked at the yards contributed for each player from September to December, usually when the regular season is held. The yards contributed are defined as the total number of passing, rushing, and receiving yards. January and February were not considered as that is when playoff games are usually held. After getting each player's yards contributed, we subsetting the data into months, took the average of each amount within that month, and outputted the results in a bar plot.

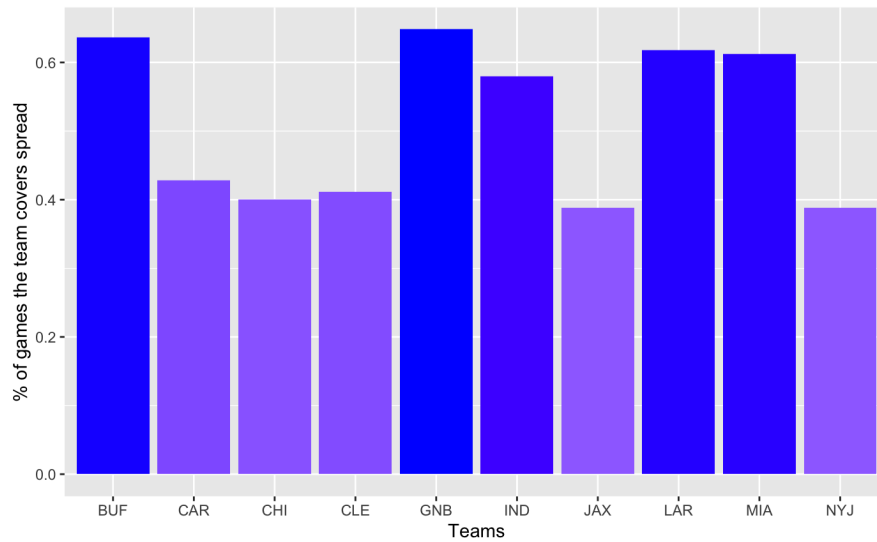
However, the yards contributed from each player does not tell the full story of a player's performance as players are able to get touchdowns from a varying number of distances.

Therefore, we decided to change the average fantasy points earned throughout the months of the regular season as fantasy points not only consider yards gained, but also touchdowns scored and catches made. From the bar plot above, we can see that there is a slight decrease from September to November and then minimally increases during December. This implies that players do suffer some burnout and decrease in performance as they have to play a lot of games that can last up to 3 hours. But when December comes along, players try to perform better as they want their team to reach the playoffs.

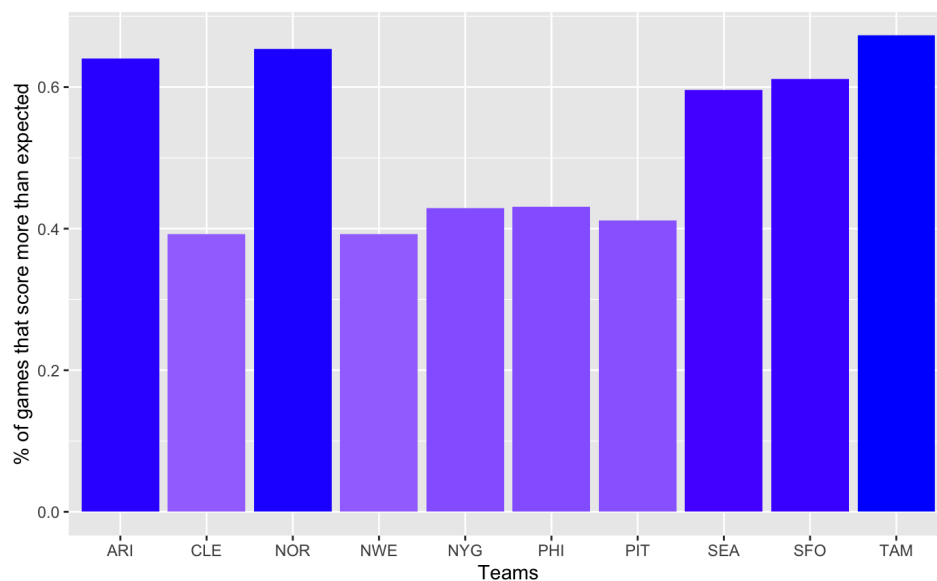
## **5. What teams are the most predictable when betting on their games?**

To find the most predictable teams, there were two metrics we looked at: If the team covered the spread and if both team's combined score was greater than the over/under. The most unpredictable teams will either cover the spread or have their games go greater than the over/under 50% of the time, while the predictable teams will be closer to either 100% or 0%.

The first metric looks at whether the underdog of the game either wins or loses by less than the given spread. For every team, we calculated the percentage of games where they covered the spread, or scored more points compared to their opponents than expected. Below shows the top and bottom 5 teams that covered the spread the most and the least.

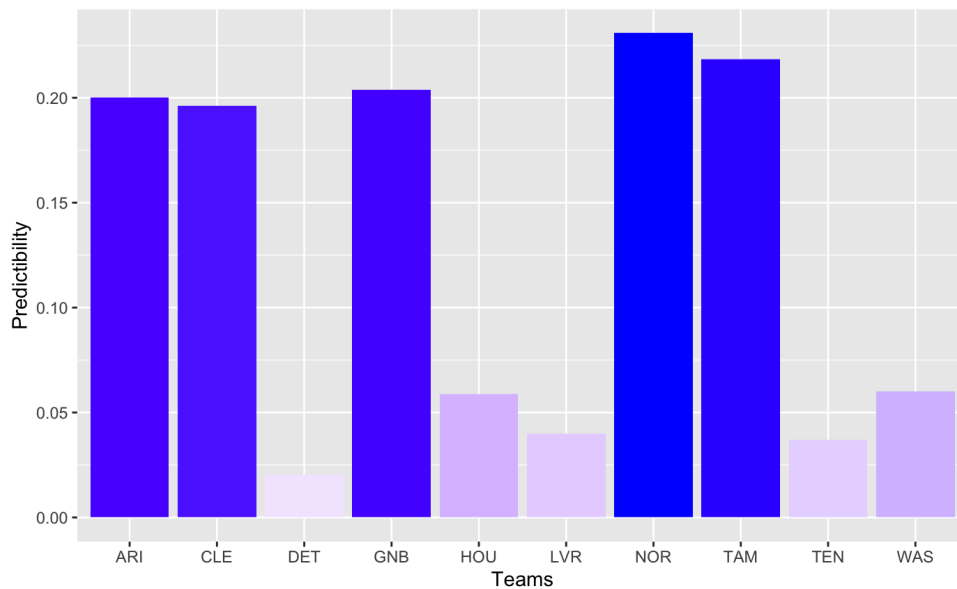


The second metric looks at whether the combined score of the team and their opponent is above or below the over/under, the expected value of their combined score. Similarly, we calculated the percentage of games where both teams scored more than expected. Below shows the top and bottom 5 teams whose games scored more than expected the most and the least.



With both metrics, the further away from 50%, the less unexpected a team will play. We found that there are no teams that are in the top or bottom 5 for both metrics. To find the

most predictable teams, we found the differences between both of their metrics and 50%, then added them up.



#### Most Predictable:

1. New Orleans Saints - 23%
2. Tampa Bay Buccaneers - 21.8%
3. Green Bay Packers - 20.3%
4. Arizona Cardinals - 20%
5. Cleveland Browns - 19.6%

#### Least Predictable:

1. Detroit Lions - 2.0%
2. Tennessee Titans - 3.7%
3. Las Vegas Raiders - 4.0%
4. Houston Texans - 5.8%
5. Washington Redskins - 6.0%

In conclusion, even the most predictable teams are unpredictable, with the most predictable team only being 23% predictable. This shows the volume of variety a team's performance can have in the NFL.

## Dead End Questions

### 6. How much value is the 1st overall pick worth compared to the other picks?

At first glance, this question seemed feasible to analyze. However, upon our first look at the data, we found that there were a significant number of players that had the same name; about 100 of them. As a result, we would have had to manually distinguish each individual player in our main dataset from each other and find the matching ID for both datasets. We decided to instead use our limited time to focus on other questions.

### 7. Does the physical attributes of a player (weight, height, etc.) affect his performance?

To answer this question, we would have to access a database for physical attributes of each player in our data set. This will be a tedious process, as the weight and height of a player can change throughout his career and also some players appear twice in the data. Therefore, we ignored this question. Given time, our prediction is that there is no correlation between physical attributes and performance as players have a great amount of muscle so that they can beat their opponents.

### 8. How does a team perform in the regular season compared to the playoffs?

At first, this question also seemed feasible to analyze. However, when we tried to process the data, we found out that we had to separate the months of the regular season and playoffs. As this was a tedious process, we decided to leave off this question. If we had more time, we could predict that a team performs in the playoffs better than in the regular season because in the playoffs, the season is at stake.

### 9. Does playing an overtime game affect the next week's performance?

We had thought that this question would have interesting results, because if a team plays in a game that goes into overtime their players may be more tired and not recover in time for the game the following week. While this sounded like an interesting question it found mild and difficult to interpret results especially when this question considers a lot more than can be reasonably shown in a single plot. We also found that it would be difficult to make multiple plots and combine their findings into a result that was interesting or easily understandable.

## Conclusions

Because of the variation of our questions, we found several interesting observations. One was the slight but significant improvement of team performance after a blowout. Another was how the fantasy points of players were skewed, with the quarterback position having the most impact on a fantasy team. We also found how much home-field advantage contributed to whether a team was a favorite to win or not. Another observation was that players tended to decrease in offensive performance towards the season, though not enough to burn out. Lastly, we found that although some teams were more predictable than others, it is hard to find a reliable team to place bets on. Though our questions were not directed towards one particular solution, we did find that there were many ways to optimize performance, whether it was in a fantasy team, as an underdog, while betting, or after a blowout.

If we further analyzed this dataset in the future, we would look at several more things. Firstly, we would look if a blowout affected a team throughout multiple games instead of one. We would also compare the changes between a blowout and the next game to the changes between two typical NFL games. We would also look at fantasy performances in the postseason compared to the regular season, since these games consist of better offenses and defense as well as higher stakes. For bets, we may look into each individual season, as teams vary a lot in between performances. Lastly, we would look at what variables could affect an underdog's chances of winning. Given the many variables we had, with more time, we could look into each one and see if that makes an underdog more or less likely to win.

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